

Community health needs assessment for a child health promotion program in Kyrgyzstan

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ARTICLE INFO

Keywords:

Needs assessment
Health promotion
Child health
Community participation

ABSTRACT

Despite over 10 years of support from government health programs and health organizations, children in migrant villages in Kyrgyzstan remain a public concern owing to high incidence rates of preventable diseases such as anemia. An effective community health promotion program is needed but there is a lack of knowledge of community barriers and health factors affecting children's health in these migrant villages. The purpose of this study was to conduct a community health needs assessment to develop strategies for a child health promotion program in these migrant villages. This study conducted using a mixed method that included literature review, regional statistics, focus group and key person interviews, and a community survey. The analysis method was descriptive statistics for the questionnaires, and content analysis for the interviews. Results showed the following priorities for child health promotion: strengthening health care provider capacity, overcoming resource shortages, managing child health and preventable diseases, improving public hygiene, promoting a healthy lifestyle, and empowering school health. The strategies were identified: capacity building of health workers, increasing community participation and raising child health awareness, and supporting child health promotion services. These findings will help guide the implementation of a child health promotion program.

1. Introduction

The people of migrant villages in Kyrgyzstan are vulnerable to health conditions and socioeconomic problems. This is particularly true in the case of preventable diseases among children such as iron-deficiency anemia and micronutrient deficiencies (Shin, 2015). The population of children in Ak-Ordo, a migrant village, was as high as 43.9% (Lenin District Office, 2016), even though the child population rate in Bishkek was 26.8% according to Kyrgyzstan statistics (National Statistical Committee of the Kyrgyzstan Republic, 2014a, 2014b). Despite migrant villages being near the capital city, people residing in these communities have limited access to social infrastructure, including medical facilities and schools.

These villages often report low levels of public support, demonstrating the need for international aid to overcome these limitations regarding the child population. Over the past 10 years, international organizations such as UNICEF, WHO, USAID, and others supported the developing country's social development as well as health state improvement. UNICEF reported that infants and toddlers have

traditionally had a serious problem of nutritional deficiency in Kyrgyzstan (UNICEF, 2014). To minimize these kinds of problems, the Ministry of Health (MoH) jointly worked with international organizations to distribute micronutrients (Serdula et al., 2013). However, the efforts of these organizations have not shown effects (Vossenaar et al., 2017) and the prevalence of preventable diseases such as child anemia is still high (WHO, 2014a, 2014b).

An effective community health promotion program is needed for children; however, there is a lack of knowledge of community barriers and health factors affecting child health in these migrant villages. To identify health problems that need prioritizing, a community health needs assessment (CHNA) is required. A CHNA is a collaborative, systematic process of collecting and analyzing data to learn about the needs of a community (Alfano-Sobsey, Ledford, Decosimo, & Horney, 2014). CHNA benefits can include better resource allocation through community partnerships, improvement in the validity of procedures, and the development of culturally appropriate health promotion intervention strategies that can positively impact health outcomes (Cottler et al., 2013).

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<https://doi.org/10.1016/j.evalprogplan.2019.02.005>

Received 21 July 2018; Received in revised form 28 December 2018; Accepted 8 February 2019

Available online 11 February 2019

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In summary, these migrant villages have a high percentage of children and face many obstacles to good health. Despite a recent increase in vulnerable community assessment and health program development (Ashwell & Barclay, 2009; Bolar et al., 2016), no studies have investigated the key health problems of Kyrgyzstan nor established a health program strategy. Assessments to plan health promotion projects in villages with a high population density of children with health problems will help to improve cost effectiveness.

This study aimed to conduct a CHNA to identify health determinants and health care patterns of the community and to develop strategies for a child health promotion program in migrant villages.

2. Methods

2.1. Study design

This was a descriptive study conducted using a mixed method that included literature review, regional statistics, focus group and key person interviews, and a community survey. After performing a multi-level assessment of community health needs and capacity, the major issues were discussed and the priorities of the program were identified.

2.2. Procedure

The CHNA procedure took place over three months. The steps in the procedure are described below (Fig. 1).

2.2.1. Composing the team

The CHNA project team for child health promotion was formed of members of a Korean NGO—the Global Korean Nursing Foundation—and received funding in 2016 to enhance capacity of international development in newly established NGOs. The research team consisted of seven researchers who were seven nursing scholars; all

were involved in the literature review but only five conducted the field survey, interview, and analysis.

2.2.2. Collection of secondary data

Before starting the fieldwork in Kyrgyzstan, the project team first conducted a literature review and planned the CHNA in advance. The project team reviewed local and national reports (e.g., from local administrative offices, public health center statistics, WHO, and websites) on health-related factors (e.g., population characteristics, disease statistics, barriers to health care). Based on this review, the project team decided to focus on CHNA for child-centered projects.

2.2.3. Collection of primary data

This study was approved by the Institutional Review Board of Kyung Hee University (KHSIRB-16-035) in Korea. The participants in this project were informed of their right to withdraw from the survey or interview at any time, and written consent forms were collected from participants. Data collection at Kyrgyzstan took place from July 21–31, 2016.

2.2.3.1. Qualitative data. Residents of migrant regions speak Kyrgyzstan, but the official language of the government is Russian, so we developed questionnaires and interview questions using Kyrgyzstan or Russian translated by professional interpreters. Interviews with government officials were conducted with an interpreter who was fluent in both Korean and Russian, and most of the interviews were conducted with an interpreter who was fluent in both Korean and Kyrgyzstan. If communication in English was possible, the interviews were conducted in English. The interviews were scheduled at a time and place most convenient for the participants (e.g., health center, office). A discussion moderator or interview host and a note taker participated in all focus group sessions and key informant interviews. Each team member conducted separate interviews; some interviews were recorded

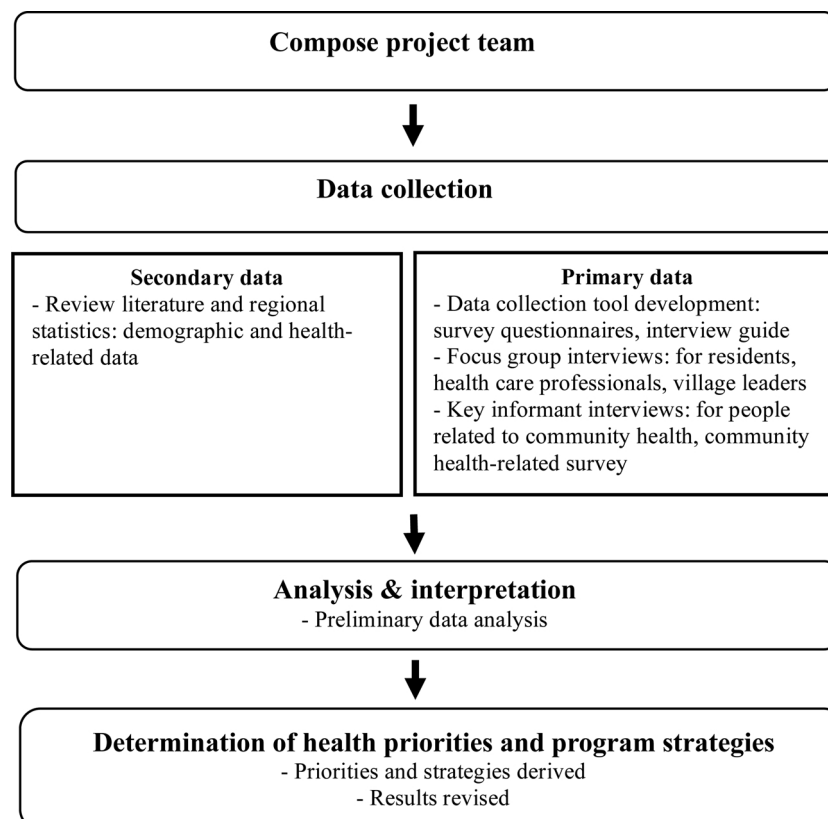


Fig. 1. Community Health Needs Assessment and Program Planning Process.

using a smartphone and later transcribed, while others were recorded by a note-taker at the interview site. Some interviewees and key persons for CHNA were introduced by the administrative director of the village and some were introduced by the chief pediatrician of the Kyrgyzstan Ministry of Health. Because the administration system in Kyrgyzstan is based on a hierarchical order, it is difficult to contact individuals, and impossible to carry out research without the permission of the person in charge. Therefore, when an unexpected key person was mentioned and needed to be included in an interview, we obtained permission from the head of his/her department and interviewed that person.

The focus group interview guide was developed based on literature review and project team discussion. The focus group interviews conducted with two groups of 16 local residents, three groups of 18 public health providers from three public health offices including nurses and doctors, and two groups of eight village leaders from three regions. The interview questions included perceptions of the health status of the community, factors affecting health (positive and negative), three things to change to improve people's health, and best and worst things about the health services. Follow-up questions and prompts were tailored as appropriate for the attendees of each meeting.

Key informant interviews were conducted with twelve key informants who were public servants and local experts, such as the chief pediatrician of the MoH in Kyrgyzstan, the chief pediatrician of department head of the city, directors of public health offices in three regions, the school president and school nurses of the two schools, and people working at international organizations such as UNICEF and USAID (SPRING team). Interview questions were about community strengths and weaknesses, barriers to health care, and needs for the health program.

2.2.3.2. Quantitative data. The sample size was determined by the population proportion formula, using a 5% margin of error, 90% confidence level, and a population of 27,000 with a child population rate of 43% in the migrant villages. The estimated sample size was 267. Considering dropouts during the study period, we chose 300 as the final sample size (Bartlett, Kotrlik, & Higgins, 2001).

Survey questionnaires were developed with input from the literature review and project team discussion; they were participants' subjective perception of 34 questions on households' socioeconomic status, health status and disease management, community environment, social support and networking, health behaviors, desired projects of a community-based program, interest in health program participation, and life satisfaction. Life satisfaction were measured using a Visual Analog Scale (VAS) with the following question: "In general, how are you satisfied with your life?". The VAS is weighted with the extremes at each end (0 = Not at all and 100 = Extremely) of a 100 mm line. Other questions were measured on a categorical scale. They were developed in Korean and translated into Kyrgyzstan, and then pilot-tested with three residents from the concerned regions, with appropriate changes made in light of their feedback. The questionnaires were distributed to the community leaders in three villages, and community leaders recruited households through convenience sampling. Adults aged 20 years or older in the households participated in the survey, and the community leader collected the surveys a week later.

2.2.4. Data analysis and interpretation

For qualitative data analysis, content analysis was done. After reading out the interview records, two researchers extracted the meaning units appeared repeatedly and then, derived the central themes through the process of sorting and reclassifying the semantic units. Through thematic analysis, original text was transformed into individual themes. Quantitative data were analyzed using SPSS software with descriptive statistics.

2.2.5. Determining health priorities and program strategies

Based on analysis of the qualitative and quantitative data, we

synthesized the priority of community health needs and derived a project strategy. After that, the priorities and strategies derived were discussed with Kyrgyzstan health officials and residents and revised for program planning within the extent applicable.

3. Results

3.1. Characteristics of health care in Kyrgyzstan (secondary data)

The health system of Kyrgyzstan is based on the Soviet model of centralized planning, hierarchical administration, and a large network of urban hospitals (World Bank, 2005). General practitioners are absent, and both efficiency and improved care are lacking (European Observatory on Health Care Systems, 2000). The healthcare policy in Kyrgyzstan (Den Sooluk program)—the National Health Insurance Reform Program (WHO, 2011)—was intended to revolutionize health care; however, the actual health care system is operating at a seriously deficient level due to insufficient workforce and poor facilities (WHO, 2014a, 2014b). The Den Sooluk program emphasizes the strengthening of the health care workforce; however, health care providers often immigrate to other countries, creating a scarcity in Kyrgyzstan. With approximately two doctors and six nurses per 1000 patients, the number of health care workers is significantly lower than in other developing countries (WHO, 2014a, 2014b). Nurses in Kyrgyzstan perform procedures and nursing care using outdated equipment and standards. Salaries are very low, and the government is not able to support the work of the nurses. The critical shortage of nurses is made worse by the lack of educational supplies and support (Pirnazarova & Schlickau, 2010). Hospitals in Kyrgyzstan are rarely equipped and care is, by Western standards, 30–50 years behind (World Bank, 2005). The rate of medical accidents is high. According to the UNICEF (2017), unreasonable pain, iatrogenic risk increase, diagnosis and treatment mismatch, wrong treatment, wrong diagnosis, and groundless hospitalizations have been observed more than 20% at 10 pilot healthcare facilities in Kyrgyzstan.

Major causes of death in children under five were perinatal complications such as prematurity, acute respiratory infections, injuries, and diarrhea (WHO, 2015). In addition, one of the major child health problems was hematologic or iron-related diseases. In Kyrgyzstan, the prevalence of childhood anemia was as high as 42% (SPRING, 2015). UNICEF (2014) reported that stunting was identified in 11.7% of children aged 6–29.9 months, wasting in 2.0%, underweight in 4.8%, and anemia in 32.7%. The prevalence of total iron deficiency was 48.0% for those with anemia.

3.2. Characteristics of health care status in migrant villages (secondary data)

This study took place in three villages, Ak-Ordo, Ak-Orgo, and Alatau, which are located in the southwestern area of Bishkek, the capital city of Kyrgyzstan. The inhabitants of this region migrated from Russia or rural areas of Kyrgyzstan in the Soviet period and the villages were formed in order of Alatau (1989 year), Ak-Orgo (1998 year), and Ak-Ordo (2005 year) after independence. The main ethnic groups in Kyrgyzstan are Kyrgyzstan (72.6%), Uzbek (14.5%), and Russian (6.4%), while the remaining 6.5% include a number of other minority groups (National Statistical Committee of the Kyrgyzstan Republic, 2014a, 2014b). In the Bishkek region, the unregistered population accounts for 30% of the population, and access to medical care is difficult (Azimov & Azimov, 2009; Sanghera et al., 2012). Furthermore, the lack of medical access has terrible consequences for unregistered residents and society (Sanghera et al., 2012).

Ak-Orgo is a newly emerging village with a relatively highly learned population and two regular schools. The schools operate from grade 1 to grade 12, and the number of students in both schools is high at around 4000. Alatau has a small population of children and a large

population of elderly people, and has one school (Shin, 2015). Finally, Ak-Ordo is the poorest village in the area; more than 90% of people in this village migrated from the countryside and there are many daily workers and high unemployment rates. Because there is no school, students must travel to neighboring Ak-Orgo village.

The residents of the migrant regions have access to a limited number of medical services because of the lack of stable income sources, frequent moves, and unregistered population.

There is one public health center in each migrant village. Providers in the public health centers are physicians, nurses, and other staff who provide medical, pediatric, and Obstetrics -Gynaecology (OB-GYN) services. After receiving primary care at the public health center, patients with serious problems are sent to #12 hospital (national general hospital), which oversees the public health center. Shin (2015) reported there were 0.97 nurses and 1.6 doctors per 1000 patients, which was a lower number of medical staffs compared to Kyrgyzstan. In addition, the most prevalent diseases in this region were cavities (51.5%) and anemia (33.8%).

3.3. Health care patterns and determinants of child health (questionnaires)

The results of the questionnaires are shown in Table 1. The questionnaires were distributed to 300 households in three villages but were returned by only 141 households (47%). They were completed by one adult per household and conducted in 22%, 51.8%, and 25.2% of Ak-Ordo, Ak-Orgo, and Alatoo villages, respectively.

Sixty-one percent of respondents were female, and the average age of respondents was 37.7 years. Overall, the economic status was poor, and some people did not eat enough. For one year, 7% of the households often lacked food for economic reasons, and 21.1% occasionally had insufficient food.

For the health status of households, 33.4% responded with “good” or “very good,” 61.3% responded with “fair,” and 5.3% responded with “bad.” The most common family diseases were anemia (48.7%) and dental caries (22.9%). In one year, 73.5% of the households visited hospitals or public health centers, while 27.7% of households said that they did not receive appropriate treatment when they were sick. In addition, only 43.5% were satisfied with medical services.

Satisfaction with the natural environment, such as air and water quality, was 74.2%, and satisfaction with living environments, including electricity, water and sewage, garbage disposal, and sports facilities, was 55.5%. These figures clearly demonstrate that the satisfaction with the environment was low.

The residents of the three villages were friendly with each other. Of the respondents, 91.5% answered that they could trust their neighbors and 82.3% answered that they helped each other with neighborhood events. The respondents did not actively engage in healthy behaviors. Many people did not pay attention to the required proportions of sweet and salty; in other words, 21.5% of households ate “very salty” and “salty” food. About 63% of people added sugar while eating or drinking, and 20.8% did not try to reduce their sugar intake. There were also very few people who exercised or underwent health checkups. Handwashing was performed by 76.2% and 86.6% of respondents before meals and after using the restroom, respectively. The results suggest the need for lifestyle modification such as diet modification, exercise, and habitual handwashing education. The life satisfaction of the villagers was relatively good, and the average life satisfaction score was high at 7.8 ± 2.6 points when evaluated on a scale of 1–10 (Table 1).

3.4. Key informants' views of health barriers and health service needs

The interviews with key informants were conducted a total of seven times. The following is a comprehensive analysis of their opinions.

- The shortage of doctors and nurses in the public health center makes it difficult to escape from an overburdened workload at the health

Table 1
General and Community Health-related Characteristics N = 141.

Category	Characteristics	%/mean (SD)
Demographic & SES	Region	
	Ak-Ordo	22.0
	Ak-Orgo	51.8
	Alatoo	26.2
	Gender (female)	61.0
	Age (mean $\pm$ SD)	37.7 (11.6)
	Occupation of household member	
	Father job (yes)	71.6
	Mother job (yes)	36.7
	Income of household	
	$\leq 10,000$ com (about \$145)	22.2
	10,001–50,000 com (about \$146–\$725)	30.0
	50,001–100,000 com (about \$726–\$1450)	20.1
	$\geq 100,001$ com (about \$1451)	27.7
	Dietary situation for 1 year	
	Sufficient amount and variety of food	42.2
	Sufficient amount but not variety of food	29.7
	Sometimes lack food because of poverty	21.1
Household health status	Often lack food because of poverty	7.0
	Single parent (Yes)	21.4
	Use public transportation without a car	59.7
	Perception of family health status	
	Very good	2.3
	Good	31.1
	Fair	61.3
	Poor	5.3
	Family members' disease	
	Tuberculosis	8.6
	Anemia	48.7
	Hepatitis	5.2
	Dental caries	22.9
	Diabetes Mellitus	3.8
	Hypertension	8.6
	Disability	3.4
	Using hospital or public health center for the past year (yes)	73.5
Community environment in the village	Sufficient to treat the disease (insufficiency)	27.7
	Satisfied with the medical service (yes)	43.5
	Water and sanitation	
	Using water supply (yes)	89.2
	Traditional toilet outside the house (yes)	89.0
	Satisfied with the safety level (yes)	41.5
	Satisfied with the natural environment (yes)	74.2
	Satisfied with the living environment (yes)	55.5
	Satisfied with public transport (yes)	59.7
	People in the village can trust each other (yes).	91.5
Social support and network	Residents help each other when there is an event in the neighborhood (yes)	82.3
	Frequency of contact with friendly neighbors	
	Less than 2–3 times a month	33.4
	Once a week	8.7
	More than 2–3 times a week	57.9
	Salt consumption (very salty and salty)	21.5
	Habit of adding salt at mealtimes (always and often)	15.4
	Efforts toward a low salt diet (never)	42.5
	Habit of adding sugar at mealtimes or tea time (always and often)	63.0
	Efforts toward a low sugar diet (never)	20.8
Health behaviors	Exercise (never)	66.2
	Health checkup in the past year (yes)	57.1

(continued on next page)

Table 1 (continued)

Category	Characteristics	%/mean (SD)
Desired program	Wash hands before meals (always)	76.2
	Wash hands after using restroom (always)	86.6
	Drinking alcohol (yes)	16.5
	Smoking (yes)	28.2
	Stress perception (much and very much)	12.9
	Health program for children in the village	
	Facilities: Playground	44.5
	Kindergarten	65.8
	School	22.6
	Hospital	30.1
Interest in health program participation	Public health center	4.1
	Education	
	Child care	6.2
	Disease management	6.2
	Intention to participate in health projects	
	Active involvement	31.1
Life satisfaction	When it comes to the situation	63.0
	Almost no participation	3.7
	No participation at all	2.2
		7.8 (2.6)

*Missing value was not included.

center. In addition, because of the lack of skills, they are unable to provide the necessary medical care. However, healthcare providers desire necessary training and wish to treat their patients appropriately.

- Unstable incomes, lack of resources such as clean water and sanitation systems, lack of educational institutions, and lack of student facilities were serious issues.
- Community residents were generally lacking in awareness on child health. Often, they did not know what their child's health condition was or how they could improve it. In addition to malnutrition, the numbers of children from single-parent families and those facing abuse were increasing.
- If the school health project is implemented, the impact of health education is likely to be significant because of numerous siblings. In addition, although the facilities in schools are scarce, the project may prove efficient because of the enthusiasm of the teachers in charge.
- The most prominent diseases among children are preventable ones such as malnutrition, anemia, and cavities. Therefore, first and foremost, the program should focus on preventable diseases for children in these villages.

3.5. People's view of health determinants and health service needs (focus group interviews)

Focus group interviews were conducted for seven groups—a total of 42 people—and revealed their views of health determinants and health service, are as follows:

3.5.1. Lack of competency of health care providers

- Still lacking in competency despite support from international organizations: The MoH has collaborated with international organizations such as USAID for over 10 years to improve child health status in Kyrgyzstan. However, the competency of nurses and physicians is low because of poor training and limited standardized educational support.
- Limited nursing education: A local nursing college, the oldest nursing school in Kyrgyzstan, and a three-year diploma program do not address the latest trends in nursing education. Most of the faculty

are physicians, and there is an obvious need for strong nursing curriculum development and capacity building.

- Training needs: The health providers from public health offices have shown a strong interest in understanding advanced health services and the use of advanced medical equipment.

3.5.2. Lack of resources

- Limited medical manpower: Most of the health services in the regional health office follow MoH guidelines, with a family physician covering over 300 community residents monthly in the form of office or home visits. As there are too many homes to visit, physicians and nurses cannot follow the mandatory home visit rules, resulting in infrequent visits.
- Difficulty in accessing health care services: When a health problem occurs, residents should visit the general hospital, which is 7 km away from the public health center; however, it is not easy to get to because of inconvenient transportation. In addition, these services are provided only to registered individuals, so unregistered people have limited access to health services.
- Limited medical equipment: Even though the main function of the public health center is to cover all aspects of the communities' overall health, family doctors and family nurses cannot give appropriate care because of limited medical equipment such as scales, oxygen supply equipment, ultrasonography, X-rays, and thermometers.

3.5.3. Insufficient safety management for children

- Child health risks: There are three primary schools (#77, #84, #82) covering first to 12th grade, and the total number of students is 6000. In these regions, there are no public bus systems or shuttle services, and over half the children commute to school by foot. Schools conduct classes in three shifts—morning, afternoon, and evening. Therefore, students are in school for limited periods, and with no pre- or afterschool programs, there are high rates of students wandering outside.

3.5.4. Passion for school health education

- School specialists' interest in health promotion education: School principals, nurses, and social workers showed a lot of interest in community health projects in the form of school health education and health scout activities.

3.5.5. High frequency of preventable diseases in children

- Since the residents and health workers are well aware of the seriousness of anemia and preventable disease in these children, they are trying to solve it, but the causes are diverse and preventive policies are not being properly implemented because of their poverty.

"There are a lot of children who could not go out because of anemia, so we (health workers) hope this project will perform well."

"the cause is maternal anemia and poor supply of iron-rich food. In Kyrgyzstan, there is a habit of drinking tea after meals. We mix baby's milk with tea. Residents later found out that it is dangerous for anemia."

"Turkey had given us anemia prevention medicine, but somebody gave it chickens."

"The residents are poor, and the environment is polluted. The children in this area are more likely to have anemia or disease than the rural area."

"Anemia drugs are expensive for poor people, and children with severe anemia cannot take them until they can buy medicines at a

cheaper price.”

3.5.6. Motivation to participate in community health programs

- Residents’ needs and will:

“I would like to actively cooperate the child health programs as a mom with two kids under five.”

“There used to be activists (CHWs) a few years ago, but activity was not good. Mothers will actively participate if they newly revitalize the CHWs.

“We want to participate in the program for my child and for my village children. If you teach us something, we would gladly follow your direction.”

- Public officials’ needs: Public officials in the region demonstrated that there is a strong need for community-based preventive health promotion services, including local governmental health programs, in the coming years.
- Public health providers’ needs and will: Of the three centers, only Ak-Ordo health center was willing to participate actively.

Ak-Ordo public health center: “Program seems to be good, but we do not have enough nurses and doctors, so our health center is hard to participate.”

Ak-Ordo public health center: “Even our situation is difficult now, we (doctors and nurses in the center) want to actively cooperate with your project. When you do the program, will you teach us the skills? Despite the desire to participate hard, we need time to learn and adapt the project. How about starting with a small number of children at first?”

3.6. Priorities and strategies for child health promotion program

Through the CHNA process, we understood many barriers of child health in the migrant community and discovered the need for a child health promotion program. Although three migrant villages initially displayed interest, only one village, Ak-Ordo, actively showed interest in participating in the project in the end. Stakeholders who understood these barriers and needs and residents of Ak-Ordo were very motivated to participate in the project. Therefore, it would be efficient to conduct a pilot child health program mainly in this region.

Based on a synthesis of the primary and secondary data, project team members discussed priorities and strategies of the areas to be addressed in the child health promotion program and revised these with Kyrgyzstan health officials and residents (Fig. 2). The six priorities identified from the data were strengthening health care provider capacity, overcoming resource shortages, managing child health and preventable diseases, improving public hygiene, lifestyle modification, and empowering school health. As a result of the project team’s discussion of strategies, it was identified that in order to manage child health in the village, capacity building of human resources was most important. The project, therefore, identified the first strategy as strengthening the capabilities of health care providers in terms of training of family doctors and nurses and provision of printed modules and screening equipment. We next focused on the use of human resources such as community residents, especially children’s parents, and school health teachers. Increasing community participation and raising child health awareness was the second strategy for the villagers themselves to recognize how to manage child health beyond the provision of resources and to maintain child health independently and continuously. Finally, we set up a strategy to support preventive health services directly between health care providers and community residents such as the support of collaboration between nurses and CHW and the support of ferrous vouchers for anemic children.

4. Discussion

4.1. Health determinants and health care patterns

This study is the first to report on CHNA with the aim of identifying determinants of health and health care patterns to develop strategies to address a child health promotion program in a migrant community in Kyrgyzstan. Social determinants of health have been positioned as central precursors to understanding and developing interventions that address root causes and advance health equity (Centers for Disease Control & Prevention, 2011). Modern/recent migration after independence of Commonwealth of Independent State (CIS) countries brought serious issues including health problems and health disparities in the population. Although the cause of anemia in this region has not been clearly elucidated, some studies have suggested that it might be associated with children in Kyrgyzstan being exposed to higher lead levels (Deryabina & El-Sadr, 2017). In this survey, unhealthy food habits such as child’s tea drinking and over-consumption of sugar appeared to be common in the village. Although these may affect childhood anemia (Fan, 2016; Temme & Van Hoydonck, 2002) and other preventable diseases, some of the interviewees did not know about the potential harm.

In this study, the ratio of visits to public health centers or hospitals was 73.5%, but only 27.7% answered that they received enough treatment and only 43.5% were satisfied with the medical service. The survey respondents included only registered residents as non-registered residents could not receive public health care services. In addition, encounters with medical personnel were not infrequent but adequate management from health care professionals was difficult because most could not afford fees for diagnostic exams or medications (most needed to pay 50% of the cost of essential medications).

Residents also listed resources that the community could use to overcome some of the barriers related to health and education. Especially, because they knew the CHNA team was planning to develop a child health promotion program, they wanted to install facilities such as a kindergarten and playground. Willingness to improve children’s health through education and lifestyle change was weak. Several years ago, there was an international development program that distributed a nutritional powder, and residents were expecting the program to be carried out the same way. Most households with low monthly income and lack of meals are in need of institutional support; however, just as in previous programs, providing a food substance is not a fundamental solution. The CHNA team once again agreed that it was important for them to improve their awareness and knowledge.

4.2. Priorities and strategies for child health promotion program

The results of CHNA are used to provide information on the topic of health promotion programs in poor areas as part of international development. To promote child health in the context of resource shortages, it is most important that doctors and nurses of health centers be able to do their job well. They need the ability to treat and prevent children’s illnesses in health centers and educate their families to prevent diseases. However, they complained that there were few basic health measurement tools in the public health center. Even though anemia was a major health problem for children in this village, there was no simple anemia test, nor did they know how to measure it and how to interpret the results. The supply of material resources cannot solve everything, but fundamental materials are essential. Assessment tools and training manuals are vital resources for health care providers. Therefore, improving health care providers’ skills through training and provision of essential tools can lead to ongoing child health care, even despite a lack of resources. Therefore, the first strategy was to build capacity for health workers at the primary level of health care.

The second strategy was to increase community participation and improve child health awareness. Community participation involves

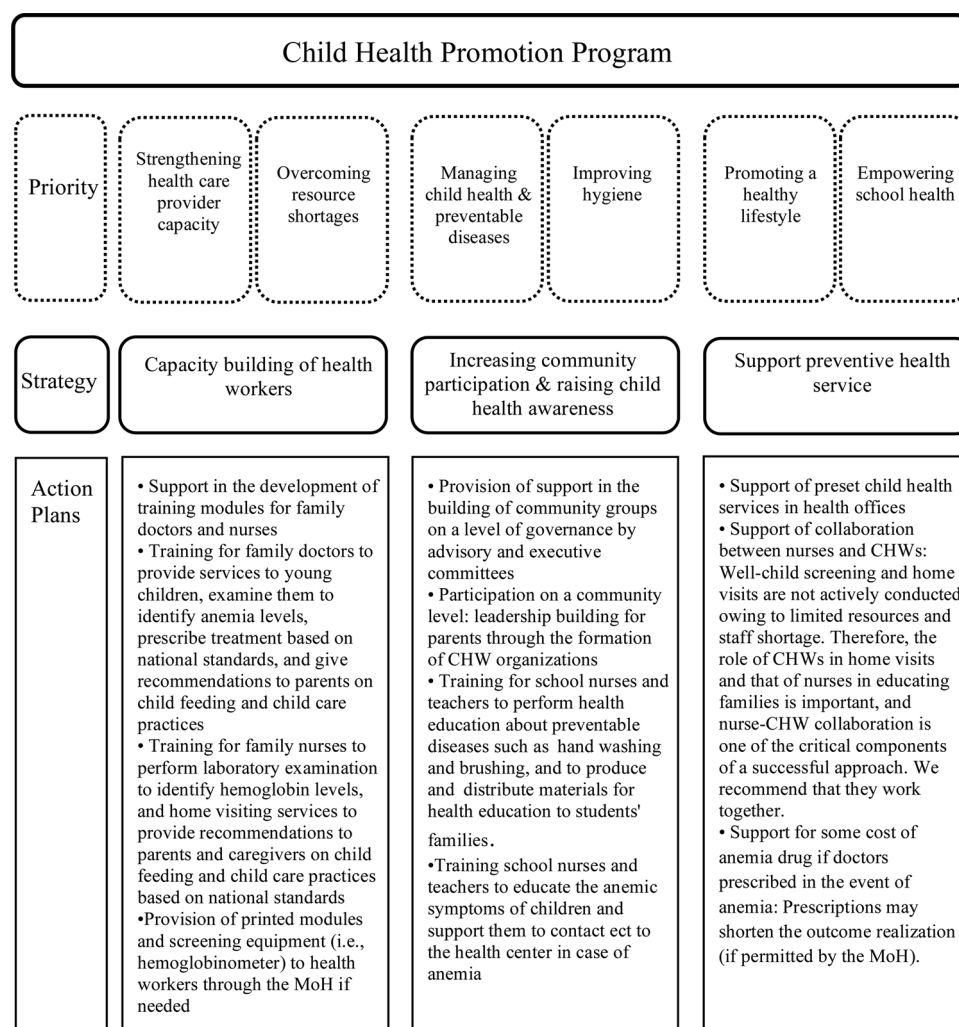


Fig. 2. Child Health Promotion Program.

engaging in health-related events and carrying out healthy life for children with the neighbors according to health workers direction and health education. Already, cases have been reported in which health and resident awareness were improved by a community participatory health program (Bolar et al., 2016; Ingram et al., 2012), while improved community awareness leads to community participation (Engle et al., 2011).

Behavior and attitude changes were required of individuals and the community. The participants stated that the community needed to find ways to contribute to child health improvement. The strength of this region is that it has good social relationship, friendly neighborhoods, and high motivation for child health care by health center providers and school teachers. It is necessary to plan the project considering these strengths of this region.

Strategies to encourage the active participation of residents are important in the prevention and management of children's illness by utilizing the characteristics of the region. Overall, 31.1% of residents in the area said that they would actively participate in the child health project, and 63% said that they would participate according to the situation. It is desirable to form a team based on the residents' intention to participate actively. The team, comprising CHW or activists, would be trained to lead individuals toward a healthy life.

Previous studies have shown that research conducted with community members increases community confidence and empowerment, creating more effective community-owned health interventions in health promotion (Wallerstein & Duran, 2010). In addition, it was

considered important for residents to form a CHW so that group-based meetings and education were conducted and monitored. While there had been a CHW group in the village before, it had shown poor results; therefore, we added a plan to activate advisory and executive committees to overcome these obstacles.

The school can promote child health awareness in the community. Based on the training by the project team, school nurses and teachers can perform health education such as hand washing and brushing and detect anemic children and contact the health center and child's parents. By disseminating the contents of school health education to the family, it is possible to promote preventive activities through public relations activities in the home of the school population. It is possible to educate and share with local residents through school and community programs to improve eating habits, which may interfere with nutrition of children, and stress the importance of handwashing.

It is effective to plan with community members rather than one-sided project teams. In this study, anemia in children was found to be serious, and mothers were more willing to collaborate to find solutions more than anyone else. In addition, the priorities and strategies determined by the researchers reflected the opinions of the community, taking the project in a more realistic direction.

The third strategy is to support child health promotion services. If the first and second strategies are to cultivate the capacity of the workforce to be placed into the child health promotion project, the third strategy is to help health workers and CHW carry out preventive health care directly. Health workers need to take care of the child

appropriate and it is necessary to support anemia medicine cost if anemia is found for the poor. The healthcare providers in the public health center and the CHW can cooperate to promote the health of the children in the community. Community residents can receive parenting advice, early screening, information on child health issues, and solutions to health problems through activities performed by health workers and CHW. Most people stated that owing to limited resources, it was difficult for them to support their children's proper growth and development. However, relying solely on an external support may be a major barrier in supporting children. Therefore, a participatory approach by the local residents is needed in child health care and child rearing.

5. Limitations

One limitation of our study was that our sample was relatively small and from a single location. Initially, the planned number of participants was 300, and the village representative distributed 300 questionnaires; however, only 141 were included in the final analysis. More than 50% were not returned, indicating a possible selection bias. It is unclear whether the unreturned surveys were similar to the returned surveys. We cannot generalize to the population from our sample. However, our goal was to collect a variety of data for health promotion projects in one village, and the possibility of generalization was not a key factor in this data collection. In addition, at first, all three migrant villages showed a keen interest in child health promotion; however, at the end of the CHNA, only the public health center of Ak-Ordo showed an active interest. The survey respondents in the area only accounted for 22% of 141 respondents. Still, we were able to supplement the information through qualitative data. Finally, most of the respondents were in their 30s, and most of the survey results seemed to reflect the opinions of young parents of school-aged children. Obtaining the opinions of family members with children of various ages would allow deeper analysis of the data.

6. Lessons learned

The first lesson learned through this study was that when planning community-based programs, it is necessary to listen to and reflect on the true voice of the health providers and community residents. Sometimes, the difficulties that arise in the health sector and the community are not covered in the secondary data or survey. In this project, we conducted qualitative data collection with questionnaires to identify local residents' health problems and their needs. We believe that in addition to the questionnaire, the qualitative interviews with community members provided additional perspectives on the child health promotion program. Integration of voices from community members enhances knowledge gained and the potential for future health promotion activities (Ainsworth, Diaz, & Schmidtlein, 2013). A quantitative and qualitative overall assessment of the health status of a village is meaningful in developing health projects (Jackson et al., 2018). Especially because it is an area where human and material resources are insufficient, cost-effective approaches are needed to address both supply and demand barriers in this vulnerable environment.

The second lesson was that flexibility is needed when project planning reflects a difference between culture and system. We did not know from the literature review but realized from the fieldwork that this village had a strong hierarchical structure. In this region, it was difficult for residents to form a team, and health-related public institutions were supposed to control them. In other words, because they were used to the "top-down" approach, they did not understand the "bottom-up" style community-based program. The "bottom-up" process positively influenced the level of cooperation and collaboration in implementing the project's community activities from participants (Green & Kreuter, 2005). Therefore, we should accept the system and culture of this region, but also show flexibility in the composition of the committee for the efficiency of the project. Thus, it is necessary to consider

the hierarchy and community-based methods at the same time when planning the pilot project. For example, the executive committee may consist mainly of those who work on the front line, but the advisory committee needs to bring bureaucracy and real practitioners together.

7. Conclusion

This study provided a comprehensive view on developing a child health promotion program in migrant communities through secondary, quantitative, and qualitative data assessments. Based on the CHNA results, we identified areas that should be priorities and strategies in a community child health promotion program. This migrant community had insufficient resources. Strategies to overcome this and to improve the health of the child were to strengthen the capacity of health care providers, encourage community participation, and support health services. Through this process, basic work has been done to develop culturally appropriate programs and further research and implementation are needed to promote child health in accordance with the strategies presented.

Funding sources

This research was supported by Kyung Hee University (KHU #20151257).

Conflict of interest

The author declared no conflict of interest.

Acknowledgments

The authors thank all members of the 2016 Monitoring and Evaluation team for the pilot program of Global Korean Nursing Foundation (GKNF).

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